Mizar Evo

Readable, AI-Ready, Scalable Formal Mathematics

Eight problems, eight proposals

A discussion with the Bialystok Mizar team, September 2026

presenter notes edition

Mizar Evo project

September 2026

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Presenter script:

- Thank the team.
- Explain that this is a design review. The team knows best whether the project still looks and reads like Mizar.
- The design is still open to change. We want to hear objections.

0.1 - A First Look

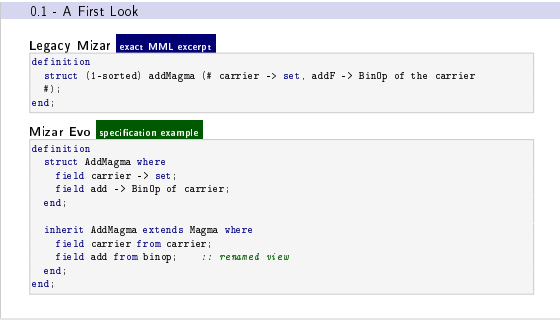
Legacy Mizar exact MML excerpt

```
definition
  struct (1-sorted) addMagma (# carrier -> set, addF -> BinOp of the carrier
  #);
end;
```

Mizar Evo specification example

```
definition
  struct AddMagma where
    field carrier -> set;
    field add -> BinOp of carrier;
  end;

  inherit AddMagma extends Magma where
    field carrier from carrier;
    field add from binop;     :: renamed view
  end;
end;
```



Presenter script: Say:

- Source: current MML `algstr_0.miz`, lines 37-40.
- This slide shows the main idea. It still reads as Mizar, and the mathematics is unchanged. The parent link and field mapping were implicit. Now they are visible in source text that the verifier can check.

0.2 - The Proposal In One Sentence

Key Phrase:

Preserve Mizar's mathematical vernacular.
Update the compiler, verifier, artifact, and publication layers.

What this talk does not claim:

- Full MML migration is not complete.
- The language standard is not final.
- AI assistance is not a substitute for proof checking.
- We build on what the current Mizar system has achieved.

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Presenter script: Read aloud:

- Full MML migration is not complete.
- The language standard is not final.
- AI assistance is not a substitute for proof checking.

After the example, explain which design obligation the syntax shows.

Every code example is labeled:

Label	Meaning
exact MML excerpt	exact current MML text, with article and line numbers
specification example	from the Mizar Evo language specification
sketch	an example; not yet fixed by the specification

Reading rule:

- No EBNF appears in this talk. The language is shown through examples only; `doc/spec/en/` remains the authority for grammar and edge cases.

deep dive

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For the table, read the row labels first, then the design reason.

0.3 - How To Read The Examples
deep dive

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0.4 - What We Need From This Visit

Key Points:

- find compatibility constraints that are hard to see outside MML work;
- choose migration benchmark articles that are small but representative;
- review the trust boundary for ATP search and kernel checking;
- discuss how Formalized Mathematics should link to a package library;
- collect the objections we have not thought of.

Main question:

What must Mizar Evo preserve so that the Mizar community still recognizes it as Mizar?

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1.1 - What Mizar Got Right

Key Points:

- declarative proof text that reads as mathematics;
- soft types, modes, and adjective-rich vocabulary;
- attributes, registrations, and clusters as reusable automation;
- a well-established, carefully maintained library (MML 5.94.1493: 1493 articles);
- a tradition of publishing formal articles (Formalized Mathematics).

Message:

- We build on these strengths. We judge every proposal by whether it protects them.

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Presenter script: Say:

- The audience knows its own system. Spend one minute on the strengths we agree on.

Read aloud:

- declarative proof text that reads as mathematics;
- soft types, modes, and adjective-rich vocabulary;
- attributes, registrations, and clusters as reusable automation;

1.2 - Pressure One: Scale

Key Points:

- MML has grown to roughly 1500 articles that depend on each other.
- The unit of dependency, review, and reuse is the whole article.
- Tools resolve article environments. But reviewers cannot see the resolved dependencies in the source.
- Whole-library maintenance operations (renames, refactorings, revisions) become more risky as the library grows.

Message:

- The problem is not that current Mizar is wrong. It is that article-level boundaries were designed for a smaller library.

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1.3 - Pressure Two: Tooling Expectations

Key Points:

- Editors are expected to give immediate feedback, even on incomplete or broken source.
- Builds are expected to be reproducible from a manifest and lockfile.
- Reuse is expected to work through packages with versions.
- Documentation is expected to be generated, linked, and browsable.

Message:

- Common programming tools already provide these features. New users expect formal libraries to provide them too.

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Key Points:

- AI agents are already useful for search, explanation, and repair.
- They need a limited amount of structured context linked to the source, rather than a copy of the whole library.
- Their output must never decide whether a proof is valid. Verification must stay independent of how capable the assistant is.
- Readable source helps here. AI editing and retrieval work best with stable local text patterns.

Message:

- Mizar's readability makes safe AI assistance possible today.

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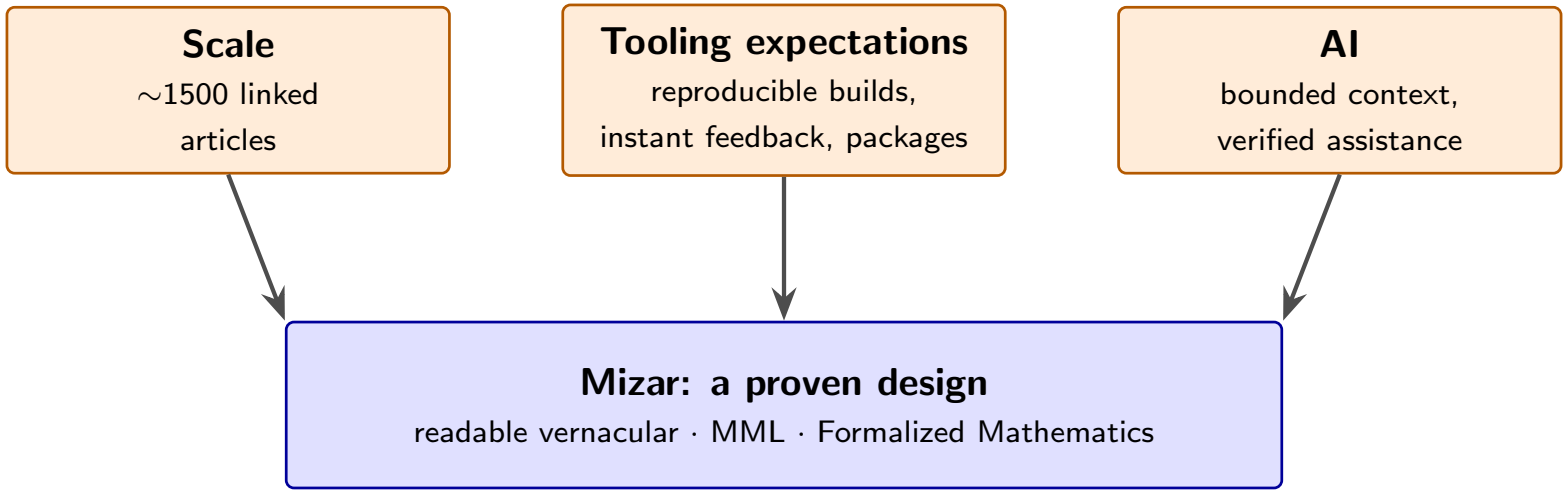
1.5 - Three Pressures, One Design



change the boundaries — keep the language

Message:

- These three needs do not mean that Mizar's design was wrong. Together, they give us reasons to change its boundaries.

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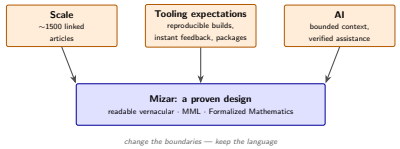

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1.6 - The Design Rule

Key Phrase:

Do not make proofs harder to read to add automation.
Use automation to protect and extend readability.

Three goals, one test:

Goal	**Test for every feature**
Readability	does proof text still read as mathematics?
AI-readiness	can a tool see a limited context that we can audit?
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Presenter script: Say:

- The eight stories that follow each apply this rule to one specific problem.

For the table, read the row labels first, then the design reason.

Legacy Mizar exact MML excerpt

```
environ

 vocabularies XBOOLE_0, SUBSET_1, BINOP_1, ZFMISC_1, STRUCT_0, ARYTM_3,
      FUNCT_1, FUNCT_5, SUPINF_2, ARYTM_1, RELAT_1, MESFUNC1, ALGSTR_0, CARD_1;
 notations TARSKI, XBOOLE_0, SUBSET_1, ZFMISC_1, BINOP_1, FUNCT_5, ORDINAL1,
      CARD_1, STRUCT_0;
 constructors BINOP_1, STRUCT_0, ZFMISC_1, FUNCT_5;
 registrations ZFMISC_1, CARD_1, STRUCT_0;
 theorems STRUCT_0;

begin :: Additive structures
```

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 theorems STRUCT_0;

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```

Presenter script: Say:

- Source: current MML `algstr_0.miz`, lines 15-25.
- Everyone here has edited these blocks, trying changes until they work.
- `environ` has supported library growth for decades. We need to consider the cost of using it with today's larger library.

2.2 - Why This Is Difficult

deep dive

Key Points:

- Information about a symbol's origin is spread across several role lists. A reviewer cannot see which article provides which notation, constructor, or cluster.
- The Accommodator resolves the environment. But the resolved dependencies are not source text that a person can review.
- Tools cannot cache or invalidate units smaller than an article.
- Moving a theorem between articles risks breaking unknown dependents.

Message:

- Implicit dependencies add work to every edit, review, and tool. That cost grows with the library.

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2.3 - The Evo Answer: Import Prelude

Mizar Evo specification example

```
import .function;
import mml.algebra.structure.sorted;

definition
  let S be 1-sorted;
  mode BinOpDef: BinOp of S is
    Function of [: S.carrier, S.carrier :], S.carrier;
end;
```

Rules that make this deterministic:

- all imports appear before the first non-import item;
- imports provide the initial active lexicon. Local declarations extend it after their declaration points; imported items retain their source FQNs;
- stable fully-qualified names are derived from package and module paths.

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After the example, explain which design obligation the syntax shows.

Mizar Evo specification example

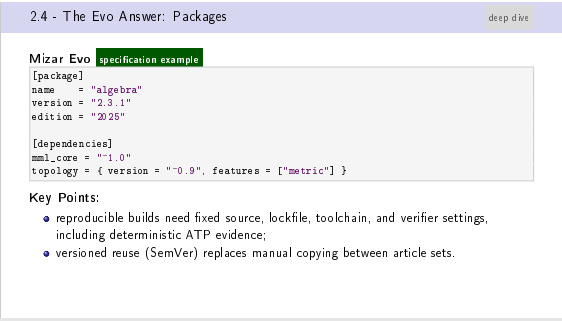
```
[package]
name    = "algebra"
version = "2.3.1"
edition = "2025"

[dependencies]
mml_core = "^1.0"
topology = { version = "^0.9", features = ["metric"] }
```

Key Points:

- reproducible builds need fixed source, lockfile, toolchain, and verifier settings, including deterministic ATP evidence;
- versioned reuse (SemVer) replaces manual copying between article sets.

deep dive



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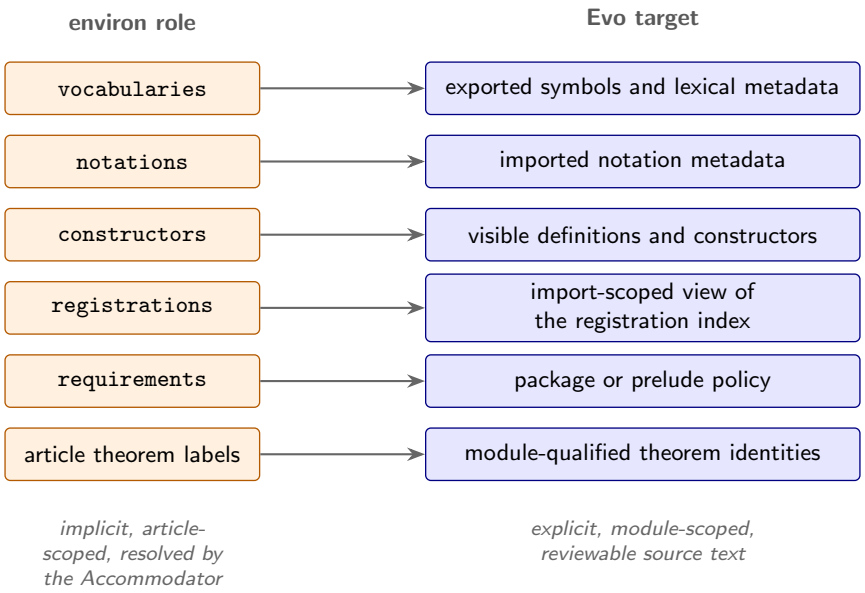
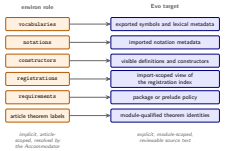
deep dive



Message:

- migration needs more than renaming. Current environments mix roles for semantics, syntax, and automation. Migration reports must explain what each imported module provides.

2.5 - Migrating The Environment
deep dive



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2.6 - What Is Preserved, What We Ask

Preserved:

- the mathematics and theorem identities are untouched;
- article-style authorship remains; a module is still a readable text;
- migration keeps origin metadata (story 8 returns to this).

Questions for Bialystok:

- Which `environ` roles must remain visually familiar during migration?
- Which current article dependencies are hardest to explain, and would make the best test cases for generated dependency reports?

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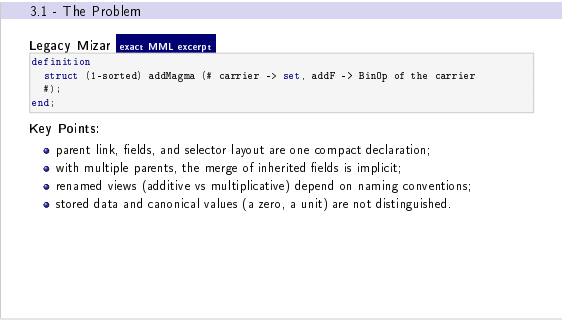
3.1 - The Problem

Legacy Mizar **exact MML excerpt**

```
definition
  struct (1-sorted) addMagma (# carrier -> set, addF -> BinOp of the carrier
  #);
end;
```

Key Points:

- parent link, fields, and selector layout are one compact declaration;
- with multiple parents, the merge of inherited fields is implicit;
- renamed views (additive vs multiplicative) depend on naming conventions;
- stored data and canonical values (a zero, a unit) are not distinguished.



Presenter script: Say:

- Source: current MML `algstr_0.miz`, lines 37-40.
- Explain why this short form was useful: structures stayed close to informal mathematical writing.

Read aloud:

- parent link, fields, and selector layout are one compact declaration;
- with multiple parents, the merge of inherited fields is implicit;
- renamed views (additive vs multiplicative) depend on naming conventions;

3.2 - Why This Is Difficult

deep dive

Key Points:

- Diamond inheritance (one structure reachable through two parent paths) is resolved by convention and declaration order, not by checkable source.
- The syntax does not tell a migration tool whether a selector is intrinsic data, a canonical value with obligations, or an inherited view.
- Errors appear far from their cause, as type mismatches in later articles.

Message:

- At MML scale, structure inheritance is a graph maintenance problem, and the graph needs explicit edges that the verifier can check.

deep dive

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3.3 - The Evo Answer: Field, Property, Attribute

Mizar Evo specification example

```
definition
  struct AddLoopStr where
    field carrier -> set;
    field add -> BinOp of carrier;
    property zero -> Element of carrier;
  end;
end;
```

Three distinct concepts:

Concept	Meaning	Consequence
`field`	stored data	constructor arguments; equality of exact instances
`property`	derived value from an implementation	`means`: existence/uniqueness; `equals`: direct term
`attribute`	predicate-style refinement	cluster propagation, not layout

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Presenter script: Say:

- A property declaration gives no value. Implementations supply values; constructors take fields only. Overlapping implementations need `coherence`.

For the table, read the row labels first, then the design reason.

3.4 - The Evo Answer: Explicit Inheritance

Mizar Evo specification example

```
definition
  struct AddMagma where
    field carrier -> set;
    field add -> BinOp of carrier;
  end;

  inherit AddMagma extends Magma where
    field carrier from carrier;
    field add from binop;     :: renamed
  end;
end;
```

- one `inherit` statement per parent; mapping and renaming are source text;
- non-identical inherited member types need `coherence` proofs of subtype inclusion. Same-name, same-type mappings need no proof;
- the additive/multiplicative naming convention becomes a checked view.

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After the example, explain which design obligation the syntax shows.

3.5 - The Evo Answer: Diamonds Become Checkable (1/2)

Mizar Evo specification example

```
definition
  struct DoubleLoopStr where
    field carrier -> set;
    field add -> BinOp of carrier;
    field mul -> BinOp of carrier;
    property zero -> Element of carrier;
    property one -> Element of carrier;
  end;

  inherit DoubleLoopStr extends AddLoopStr;
  inherit DoubleLoopStr extends MulLoopStr;
end;
```

Mizar Evo specification example

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    property one -> Element of carrier;
  end;

  inherit DoubleLoopStr extends AddLoopStr;
  inherit DoubleLoopStr extends MulLoopStr;
end;
```
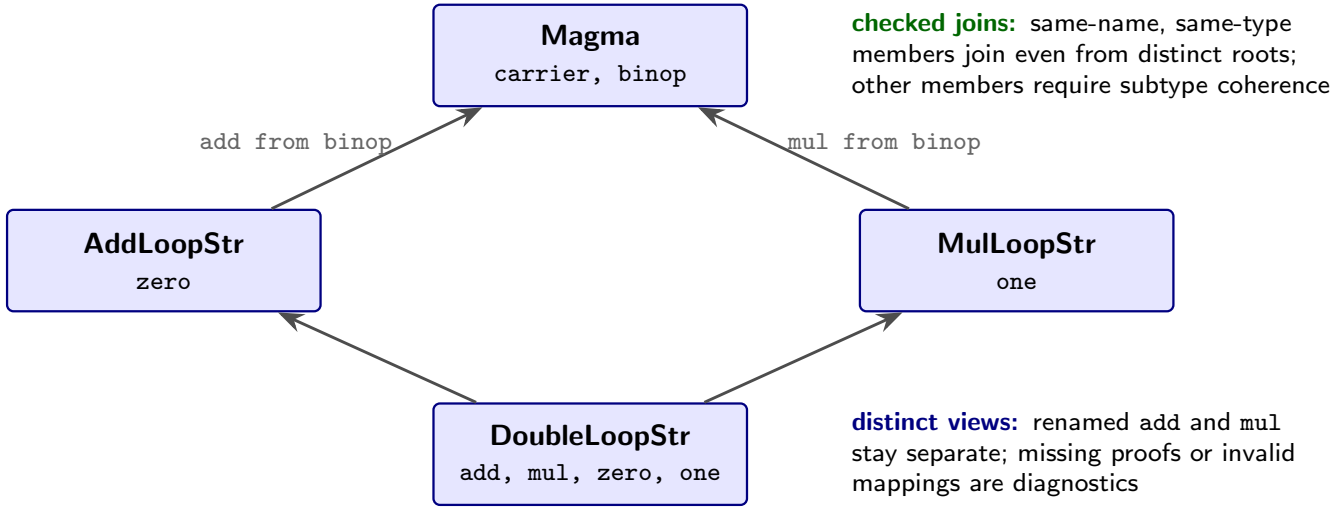

Presenter script: Say:

- The diagram assumes the shown parent mappings from `AddLoopStr` and `MulLoopStr` to `Magma`. The code adds their shared child.

3.5 - The Evo Answer: Diamonds Become Checkable (2/2)

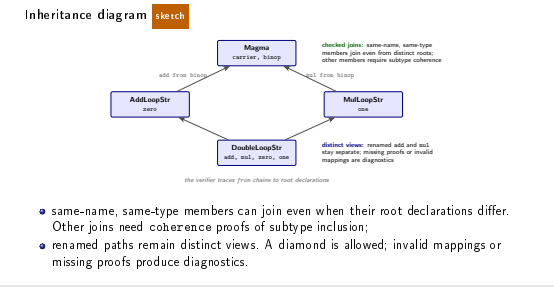
Inheritance diagram sketch



the verifier traces `from` chains to root declarations

- same-name, same-type members can join even when their root declarations differ. Other joins need `coherence` proofs of subtype inclusion;
- renamed paths remain distinct views. A diamond is allowed; invalid mappings or missing proofs produce diagnostics.

3.5 - The Evo Answer: Diamonds Become Checkable (2/2)



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- renamed paths remain distinct views. A diamond is allowed; invalid mappings or missing proofs produce diagnostics.

3.6 - What Is Preserved, What We Ask

Preserved:

- structures remain Mizar structures: carriers, selectors, `Element of`;
- aggregates become longer only where we need to make a hidden decision explicit.

Questions for Bialystok:

- Is the `field` / `property` / `attribute` split readable in real algebraic articles, or does it require too many annotations in simple cases?
- Is one-parent-per-`inherit` acceptable for parts of MML with much inheritance, such as the `ALGSTR` and topology hierarchies?
- Which MML structures would be the best diamond test cases?

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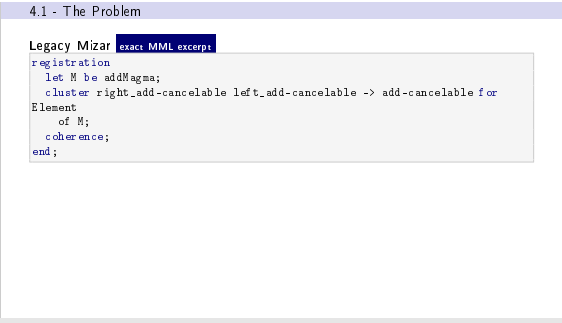
- Is one-parent-per-inherit acceptable for parts of MML with much inheritance, such as the ALGSTR and topology hierarchies?
- Which MML structures would be the best diamond test cases?

Legacy Mizar exact MML excerpt

```
registration
  let M be addMagma;
  cluster right_add-cancelable left_add-cancelable -> add-cancelable for
Element
    of M;
  coherence;
end;
```



Presenter script: Say:

- Source: current MML `algstr_0.miz`, lines 104-109.
- Registrations are one of Mizar's best ideas. Adjectives propagate automatically, so proofs stay short. But it is hard to see how this works.

4.2 - Why This Is Difficult

deep dive

Key Points:

- When a proof fails, "why does the checker not see that this is a Group?" has no local answer. The cause is somewhere in the environment.
- Users cannot see which registrations fired or in which order. The automation is powerful, but explaining it becomes harder as it does more.
- For an AI assistant the situation is worse: it must guess the cluster state instead of reading it.

Message:

- Automation without explanations makes a large library harder to maintain, even when it is sound.

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- Automation without explanations makes a large library harder to maintain, even when it is sound.

4.3 - The Evo Answer: Labeled, Traceable Registrations

Mizar Evo specification example

```
registration
  cluster EmptyImpliesFinite: empty -> finite for set;
  coherence proof ... end;

  cluster FiniteImpliesCountable: finite -> countable for set;
  coherence proof ... end;
end;
```

Key Points:

- every registration item carries a required label: citable in by, reported in diagnostics, part of the module interface;
- the verifier uses an import-filtered view of the global cluster graph;
- explain-attribute and resolution traces show why an attribute follows or why resolution fails, e.g. empty -> finite -> countable.

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After the example, explain which design obligation the syntax shows.

Mizar Evo specification example

```
registration
  let n be Nat;
  reduce NatAddZero: n + 0 to n;
  reducibility
  proof
    let n be Nat;
    thus n + 0 = n by mml.number.natural.Nat_add_zero;
  end;
end;
```

Key Points:

- a reduction is an oriented simplification backed by an equality proof;
- the right side must be strictly smaller, so imported rules cannot loop, and rule selection is deterministic (pattern subsumption, then guard specificity, then FQN tie-break);
- unoriented identification idioms become auditable `reduce` items.

deep dive

4.4 - The Evo Answer: Oriented Reductions deep dive

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4.5 - What Is Preserved, What We Ask

Preserved:

- registrations and clusters remain first-class; proofs stay short;
- cluster applications need no repeated proofs. Reductions may need local guard evidence or an explicit equality citation.

Questions for Bialystok:

- Which cluster explanations would help most with daily work: failure explanations, firing traces, or difference reports between environments?
- Which MML article families make the most complex use of registrations and should become migration benchmarks for the cluster graph?

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5.1 - The Problem

Key Points:

- Users want stronger automation: bigger by steps, hammer-style search.
- But in a monolithic verifier, every gain in search power grows the code that must be trusted.
- External provers (ATPs) are strong exactly where Mizar's core is first-order - and they are the least auditable component of all.
- MizAR and MPTP research already showed that ATP search is powerful on MML premises; the open question is trust, not power.

Key Phrase:

How do we get modern proof search
without trusting the searcher?

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Key Phrase:

How do we get modern proof search without trusting the searcher?

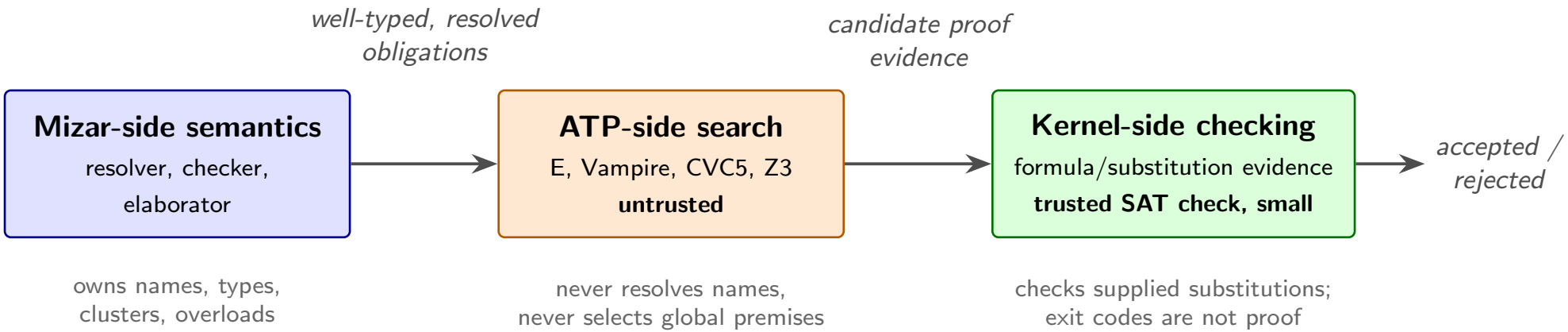
Presenter script: Say:

- Mention MizAR, MPTP, and hammer research by name. They showed how well search works on MML. Mizar Evo adds a boundary that allows us to use that search without making the trusted base larger.

Read aloud:

- Users want stronger automation: bigger by steps, hammer-style search.
- But in a monolithic verifier, every gain in search power grows the code that must be trusted.
- External provers (ATPs) are strong exactly where Mizar's core is first-order - and they are the least auditable component of all.

5.2 - The Evo Answer: A Reasoning Boundary



Key rules:

- ATPs never resolve names, infer types, expand clusters, or pick overloads;
- the kernel checks supplied formulas and substitutions, then runs its trusted SAT check. It does not select premises or invent substitutions;
- deterministic pre-ATP discharge needs replayable evidence too - nothing is accepted because an earlier phase said "done".

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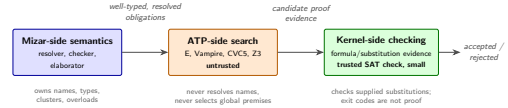

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- deterministic pre-ATP discharge needs replayable evidence too - nothing is accepted because an earlier phase said "done".

Presenter script: Say:

- Development policy may record `externally_attested` results. These are separate from kernel-verified proofs.

Read aloud:

- ATPs never resolve names, infer types, expand clusters, or pick overloads;
- the kernel checks supplied formulas and substitutions, then runs its trusted SAT check. It does not select premises or invent substitutions;
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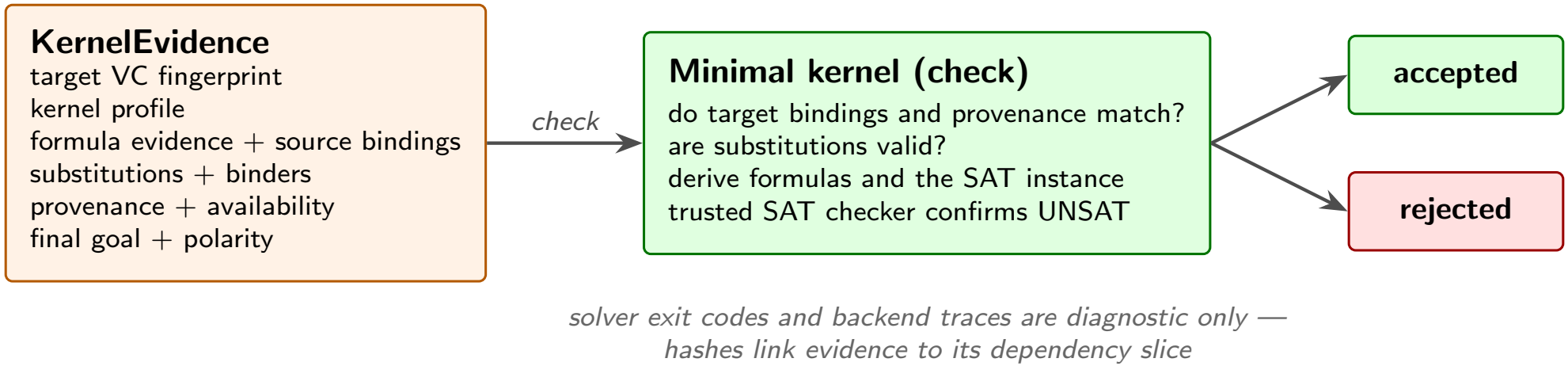
5.3 - Formula And Substitution Evidence



- `KernelEvidence` contains source formulas, explicit substitutions, provenance, and target/goal bindings;
- the kernel checks this evidence, derives instantiated formulas and SAT clauses, and requires UNSAT from its trusted in-process Rust SAT checker;
- backend resolution traces, SMT proof objects, logs, and exit codes are not trusted acceptance evidence.

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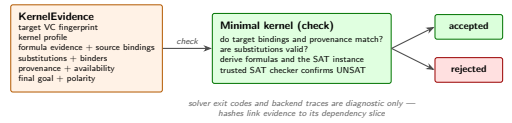

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- backend resolution traces, SMT proof objects, logs, and exit codes are not trusted acceptance evidence.

Presenter script: Say:

- Hashes bind evidence to its dependency context. The kernel also checks binder conditions and the goal's refutation polarity.

Read aloud:

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- backend resolution traces, SMT proof objects, logs, and exit codes are not trusted acceptance evidence.

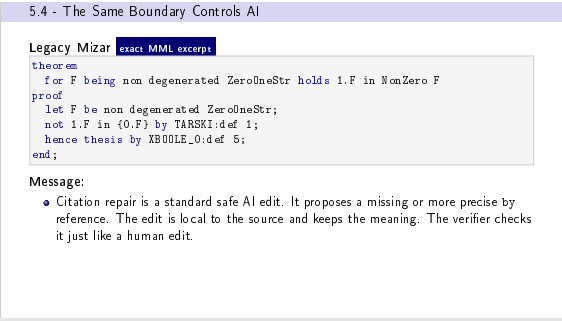
5.4 - The Same Boundary Controls AI

Legacy Mizar **exact MML excerpt**

```
theorem
  for F being non degenerated ZeroOneStr holds 1.F in NonZero F
proof
  let F be non degenerated ZeroOneStr;
  not 1.F in {0.F} by TARSKI:def 1;
  hence thesis by XBOOLE_0:def 5;
end;
```

Message:

- Citation repair is a standard safe AI edit. It proposes a missing or more precise by reference. The edit is local to the source and keeps the meaning. The verifier checks it just like a human edit.



Presenter script: Say:

- Source: current MML `struct_0.miz`, lines 637-643.
- The agent proposes; the verifier and kernel decide. The assistant's strength never enters the trusted base.

Read aloud:

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5.5 - Edit Classes: Green, Yellow, Red

deep dive

Class	Examples	Policy
Green	add citation, insert qua, info annotation	can be proposed automatically; still verified
Yellow	add import, local lemma, registration	proposed with human review
Red	weaken theorem, change definition, add axiom	forbidden to ordinary agents

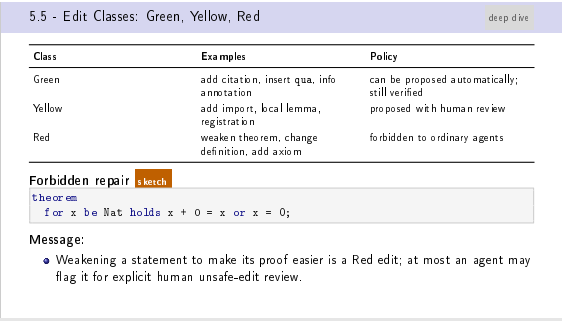
Forbidden repair sketch

```
theorem
  for x be Nat holds x + 0 = x or x = 0;
```

Message:

- Weakening a statement to make its proof easier is a Red edit; at most an agent may flag it for explicit human unsafe-edit review.

deep dive



Presenter script: Read aloud:

- Weakening a statement to make its proof easier is a Red edit; at most an agent may flag it for explicit human unsafe-edit review.

After the example, explain which design obligation the syntax shows.
For the table, read the row labels first, then the design reason.

5.6 - What Is Preserved, What We Ask

Preserved:

- the de Bruijn discipline: acceptance uses a small trusted core, including its SAT checker;
- proof text stays declarative and readable - automation maintains the argument, it does not replace it.

Questions for Bialystok:

- Is the formula/substitution evidence approach convincing for Mizar-style obligations, including clusters and definitional expansions?
- Which evidence format would the team be most willing to audit?

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6.1 - The Problem

Key Points:

- Verifying the whole MML is a batch operation measured in hours.
- The reuse boundary is the accepted article: a small change re-verifies more than it should.
- Memory follows the article environment, not the actually used interface.
- These costs follow from using the article as the unit in a large library. They are not errors in the current design.

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6.2 - The Evo Answer: Fingerprints And Incrementality (1/2)

Key Points:

- cache keys cover source, dependency slices, package/lockfile, toolchain, schema, policy, registrations, obligations, evidence, and witnesses;
- reuse requires all relevant keys to match; missing data causes a cache miss;
- a theorem proof-body edit does not rebuild importers if its exported statement and accepted status stay unchanged;
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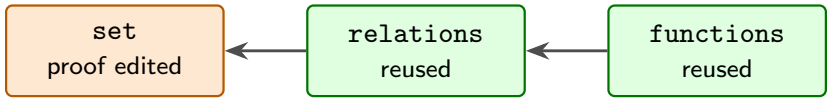
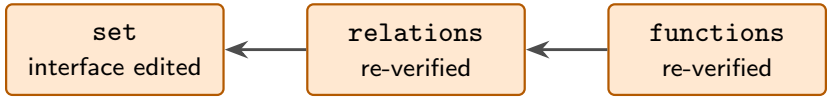
6.2 - The Evo Answer: Fingerprints And Incrementality (2/2)

theorem proof edit:
public statement + accepted status unchanged — importers reuse

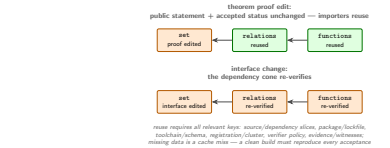


interface change:
the dependency cone re-verifies



reuse requires all relevant keys: source/dependency slices, package/lockfile, toolchain/schema, registration/cluster, verifier policy, evidence/witnesses; missing data is a cache miss — a clean build must reproduce every acceptance

Cache reuse is never proof authority.
A clean build must reproduce every acceptance.



Cache reuse is never proof authority.
A clean build must reproduce every acceptance.

Presenter script: After the example, explain which design obligation the syntax shows. Introduce the next topic: The Evo Answer: Fingerprints And Incrementality (2/2). Point to the example, question, or diagram before you continue.

6.3 - The Evo Answer: A Memory Contract

deep dive

```
resident memory should scale with:
  active source and typed AST
  imported public interfaces
  import-filtered indexes
  active module obligations
  in-flight proof checks and ATP runs
  small caches

not with:
  imported proof bodies
  imported proof witnesses
  private lemmas outside the interface
  registration data outside the import closure
```

Message:

- This is a resident-memory model, not a measured performance guarantee. Proof bodies and traces load lazily for specific queries.

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Presenter script: Read aloud:

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After the example, explain which design obligation the syntax shows.

6.4 - What Is Preserved, What We Ask

Preserved:

- clean-build semantics: caching and parallelism change speed, never truth;
- article-style review: what a human reads is still complete source text.

Questions for Bialystok:

- What clean-build equivalence tests would make incremental verification trustworthy to the team that maintains MML today?
- Which current maintenance operations (revisions, renamings) should we benchmark for incremental cost?

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7.1 - The Problem, Part One: Schemes Use Separate Rules

Legacy Mizar **exact MML excerpt**

```
scheme
  NatInd { P[Nat] } : for k being Nat holds P[k]
provided
A1: P[0] and
A2: for k be Nat st P[k] holds P[k + 1]
```

Key Points:

- schemes support second-order patterns (induction, separation, replacement). They work, but they use a separate mechanism with separate rules;
- schemes can parameterize theorems, but not structures, modes, or functors.

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Presenter script: Say:

- Source: current MML `nat_1.miz`, near line 90 (checked July 2, 2026).

Read aloud:

- schemes support second-order patterns (induction, separation, replacement). They work, but they use a separate mechanism with separate rules;
- schemes can parameterize theorems, but not structures, modes, or functors.

7.2 - The Problem, Part Two: Copy-Paste Algebra

Key Points:

- `addMagma` and `multMagma` are the same mathematics twice, related only by naming convention (we saw this in story 2);
- polynomial rings, vector spaces, and matrix theories are written again for each carrier because there is no parameterized construction;
- a theorem proved for one commutative operation is re-proved for + and $*$ separately.

Message:

- Without a generics mechanism, the library needs duplicate articles. Each copy needs maintenance.

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- Without a generics mechanism, the library needs duplicate articles. Each copy needs maintenance.

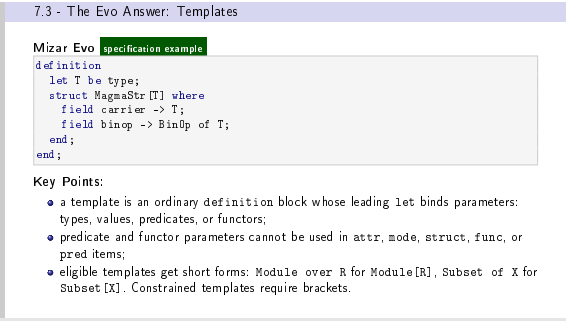
7.3 - The Evo Answer: Templates

Mizar Evo specification example

```
definition
  let T be type;
  struct MagmaStr[T] where
    field carrier -> T;
    field binop -> BinOp of T;
  end;
end;
```

Key Points:

- a template is an ordinary `definition` block whose leading `let` binds parameters: types, values, predicates, or functors;
- predicate and functor parameters cannot be used in `attr`, `mode`, `struct`, `func`, or `pred` items;
- eligible templates get short forms: `Module over R` for `Module[R]`, `Subset of X` for `Subset[X]`. Constrained templates require brackets.



Presenter script: Read aloud:

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- eligible templates get short forms: Module over R for Module[R], Subset of X for Subset[X]. Constrained templates require brackets.

After the example, explain which design obligation the syntax shows.

7.4 - Bounded Parameters And Generic Theorems

deep dive

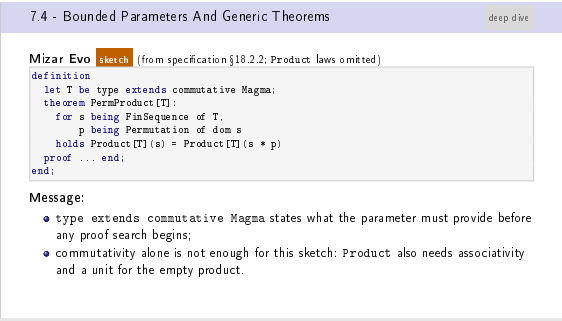
Mizar Evo sketch (from specification §18.2.2; `Product` laws omitted)

```
definition
  let T be type extends commutative Magma;
  theorem PermProduct[T]:
    for s being FinSequence of T,
        p being Permutation of dom s
    holds Product[T](s) = Product[T](s * p)
  proof ... end;
end;
```

Message:

- `type extends commutative Magma` states what the parameter must provide before any proof search begins;
- commutativity alone is not enough for this sketch: `Product` also needs associativity and a unit for the empty product.

deep dive



Presenter script: Read aloud:

- Mizar Evo (sketch from specification §18.2.2; Product laws omitted):
- type extends commutative Magma states what the parameter must provide before any proof search begins;
- commutativity alone is not enough for this sketch: Product also needs associativity and a unit for the empty product.

After the example, explain which design obligation the syntax shows.

7.5 - One Proof, Many Instantiations

deep dive

Instantiation sketch (; assumes the required laws and attribute evidence)

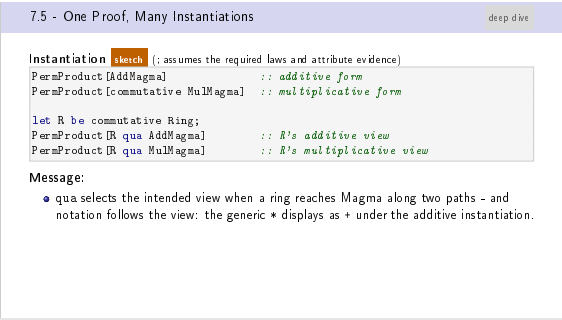
```
PermProduct[AddMagma]                 :: additive form
PermProduct[commutative MulMagma]     :: multiplicative form

let R be commutative Ring;
PermProduct[R qua AddMagma]           :: R's additive view
PermProduct[R qua MulMagma]           :: R's multiplicative view
```

Message:

- qua selects the intended view when a ring reaches Magma along two paths - and notation follows the view: the generic $*$ displays as + under the additive instantiation.

deep dive



Presenter script: Read aloud:

- Instantiation (sketch; assumes the required laws and attribute evidence):
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After the example, explain which design obligation the syntax shows.

Mizar Evo specification example

```
definition
  let P be pred(Nat);
  theorem NatInduction[P]:
    P(0) & (for n being Nat st P(n) holds P(n+1))
    implies for n being Nat holds P(n)
  proof ... end;
end;
```

Key Points:

- predicate parameters follow the familiar `defpred` convention;
- legacy schemes have a direct, mechanical migration target;
- theorem instantiation uses explicit brackets, so tools and artifacts see exactly which instance a proof uses.

deep dive

7.6 - Schemes Become Ordinary Templates deep dive

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7.7 - What Is Preserved, What We Ask

Preserved:

- scheme-style reasoning keeps the same power;
- `of` / `over` phrasing keeps mathematical prose readable;
- first-order discipline: each instantiation is checked; templates add no new logic.

Questions for Bialystok:

- Which MML schemes should be the first migration targets?
- Are brackets acceptable as the canonical identity form, with `of`/`over` as display forms?
- For `func` and `pred` templates, normalized declared argument types must determine each inferred type parameter uniquely. `qua` views are never inferred. Does this rule require too many explicit `[T]` arguments?

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8.1 - The Problem

Key Points:

- Current Mizar can define `Gcd` and prove its theory - but cannot compute `gcd(48, 18)`; every numeric fact needs a hand-written proof chain.
- There is no checked connection between MML mathematics and executable code; verified-algorithm work must leave the system entirely.
- This is a boundary of the design, not a defect: Mizar chose to be a proof language. The question is whether that boundary still meets our needs.

Key Phrase:

The library describes computation.
It cannot perform or export it.

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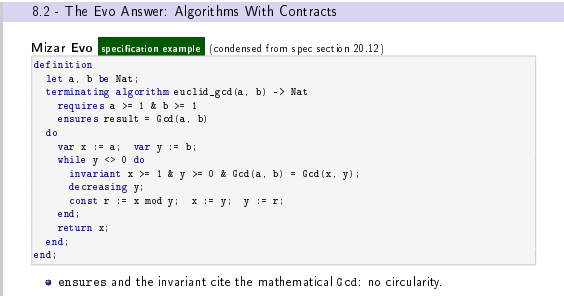
After the example, explain which design obligation the syntax shows.

8.2 - The Evo Answer: Algorithms With Contracts

Mizar Evo **specification example** (condensed from spec section 20.12)

```
definition
  let a, b be Nat;
  terminating algorithm euclid_gcd(a, b) -> Nat
    requires a >= 1 & b >= 1
    ensures result = Gcd(a, b)
  do
    var x := a;  var y := b;
    while y <> 0 do
      invariant x >= 1 & y >= 0 & Gcd(a, b) = Gcd(x, y);
      decreasing y;
      const r := x mod y;  x := y;  y := r;
    end;
    return x;
  end;
end;
```

- ensures and the invariant cite the mathematical Gcd: no circularity.



Presenter script: Say:

- The contract uses mathematics. The algorithm computes the result, and the library functor `Gcd` specifies it. The `decreasing` measure on `y` proves termination.

Read aloud:

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8.3 - The Evo Answer: Proof By Computation

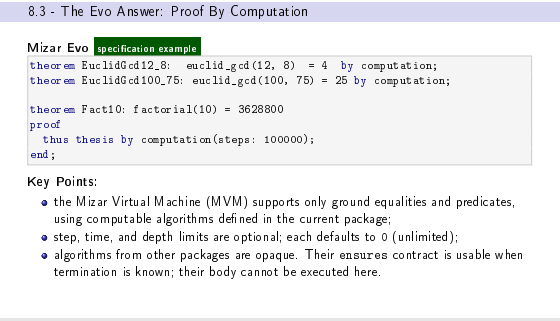
Mizar Evo specification example

```
theorem EuclidGcd12_8:  euclid_gcd(12, 8)  = 4  by computation;
theorem EuclidGcd100_75: euclid_gcd(100, 75) = 25 by computation;

theorem Fact10: factorial(10) = 3628800
proof
  thus thesis by computation(steps: 100000);
end;
```

Key Points:

- the Mizar Virtual Machine (MVM) supports only ground equalities and predicates, using computable algorithms defined in the current package;
- step, time, and depth limits are optional; each defaults to 0 (unlimited);
- algorithms from other packages are opaque. Their `ensures` contract is usable when termination is known; their body cannot be executed here.



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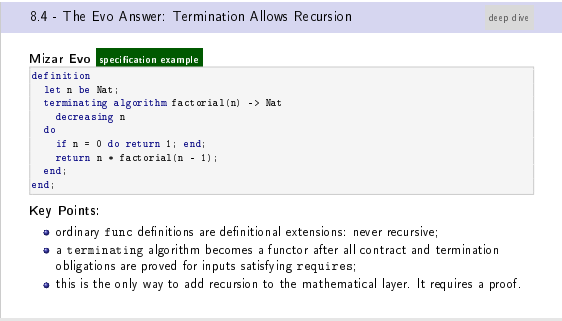
After the example, explain which design obligation the syntax shows.

Mizar Evo specification example

```
definition
  let n be Nat;
  terminating algorithm factorial(n) -> Nat
    decreasing n
  do
    if n = 0 do return 1; end;
    return n * factorial(n - 1);
  end;
end;
```

Key Points:

- ordinary `func` definitions are definitional extensions: never recursive;
- a `terminating` algorithm becomes a functor after all contract and termination obligations are proved for inputs satisfying `requires`;
- this is the only way to add recursion to the mathematical layer. It requires a proof.



Presenter script: Say:

- Each call must satisfy `requires`. An ordinary algorithm may prove only partial correctness and is not promoted to a functor.

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8.5 - Computation Never Redefines Truth

deep dive

Key Points:

- contracts, invariants, and termination measures generate verification conditions that pass through the same ATP-plus-kernel boundary as ordinary theorems (story 4);
- `by computation` uses MVM replay with optional limits;
- code extraction (to runtime targets) is strictly downstream of verified artifacts and can never affect acceptance.

Message:

- Algorithms let the library express more. They do not change what it means for a theorem to be accepted.

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8.6 - What Is Preserved, What We Ask

Preserved:

- algorithms live in `definition` blocks. Proofs can use verified promotion or `by computation` for computable ground calls in the current package;
- a partial algorithm's `ensures` needs evidence that the call terminates;
- first-order set-theoretic foundations stay intact.

Questions for Bialystok:

- Which computational examples would show clear benefits to mathematicians without shifting the culture toward programming?
- Are there MML areas (number theory, combinatorics, finite structures) where `by computation` would immediately shorten real proofs?
- Which extraction targets matter first, if any?

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9.1 - The Problem

Key Points:

- Formalized Mathematics gives Mizar a rare feature: a research journal linked to a formal library, with articles that people can cite.
- But article identity is closely tied to library organization. Refactoring the library can conflict with the structure of published articles. Package reuse has no identity for journal citations.
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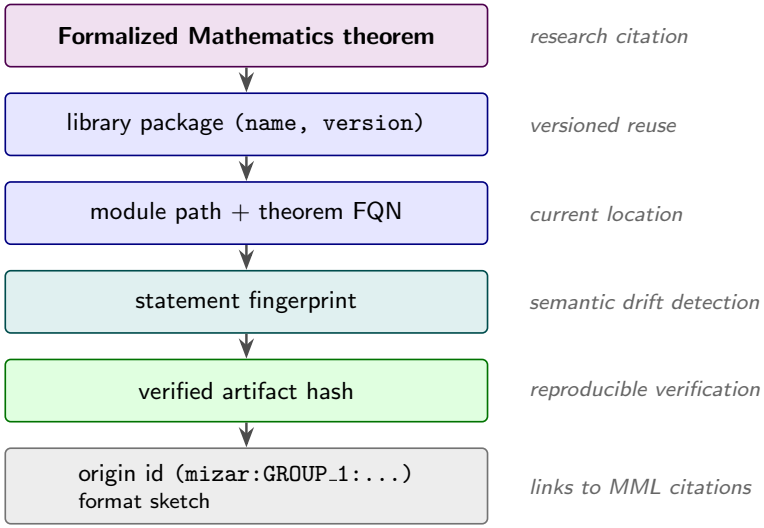
9.2 - The Evo Proposal: Linked Records



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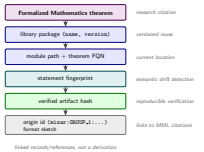

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9.3 - Who Gains What

deep dive

Audience	**Gain**
readers	written explanations come first; formal source is one click away
maintainers	refactoring no longer rewrites published explanations
authors	frozen `pub` article identities; FQNs for library locations
planned AI tools	prose for retrieval, fingerprints for exact context

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- Formalized Mathematics remains a real journal with review and written explanations;
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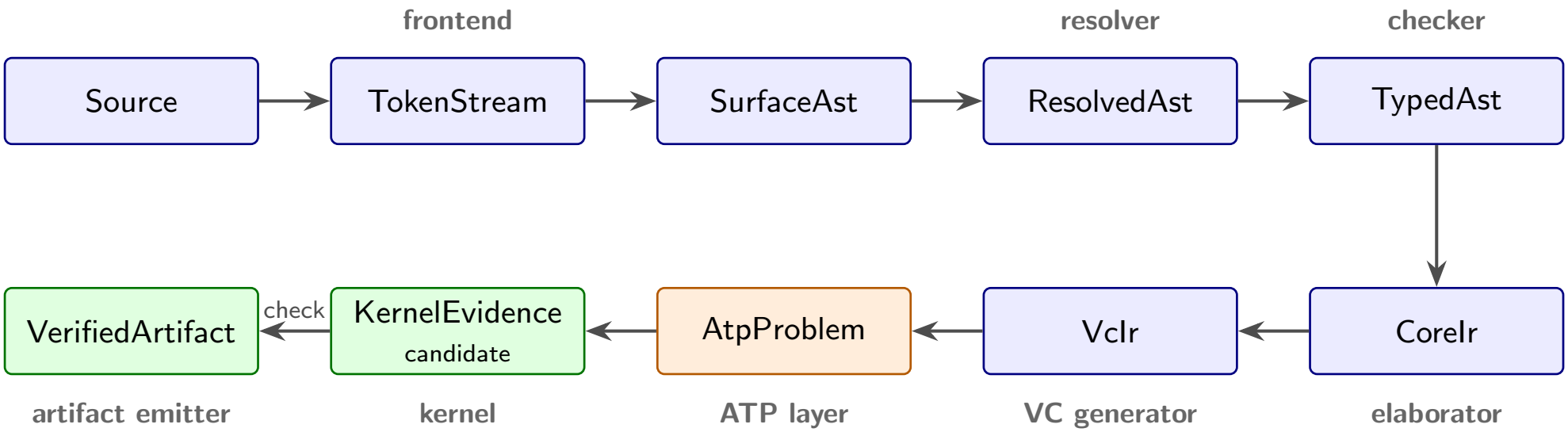

Message:

- every boundary states who owns a fact, which artifact records it, and what must be recomputed when it changes. That is the main purpose.

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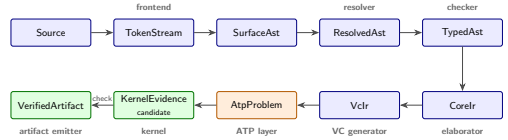

Message:

- every boundary states who owns a fact, which artifact records it, and what must be recomputed when it changes. That is the main purpose.

Presenter script: Say:

- The diagram shows the ATP path. Only obligations still open after deterministic discharge go to ATP; earlier discharge also needs evidence.

Read aloud:

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10.2 - Responsibility Split

deep dive

Layer	Responsibility
frontend	lexing, parsing, recovery
resolver	imports, names, labels, namespaces
checker	soft types, clusters, registrations, overloads
elaborator	core logical representation
VC generator	proof and algorithm obligations
ATP layer plus kernel	untrusted search, then acceptance by checking
artifact emitter	stable outputs for tools and dependents

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10.3 - Pipeline Stages For The Eight Stories

Story	Pipeline stage
dependencies	resolver, package manager
structures and automation	checker (inheritance and cluster graphs, traces)
search vs trust	ATP layer, kernel, formula/substitution evidence
scale	artifacts, fingerprints, scheduler
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Message:

- The eight stories describe one pipeline. Each starts with a different problem that users face.

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10.4 - Testing The Trust Boundary

deep dive

Key Phrase:

Reject what must not pass
before
Accept everything that should pass

Key Points:

- soundness bugs have higher priority than parser gaps. Tests near the kernel first focus on malformed and failing evidence;
- with these tests in place, we add coverage for accepted language forms.

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11.1 - Where The Project Stands Today

Key Points:

- bilingual language specification: 24 chapters plus appendices, English canonical with Japanese companions (`doc/spec/`);
- 24 draft architecture documents covering the pipeline, kernel, kernel evidence, and AI agent interface (`doc/design/architecture/`);
- a Rust workspace of 20 crates - lexer, parser, resolver, checker, VC generator, ATP bridge, kernel, build system - roughly 400k lines including tests;
- focused audits completed in 2026: kernel soundness, template logic encoding, SAT solver dependency;
- the roadmap is split into small tasks that can be verified separately.

Message:

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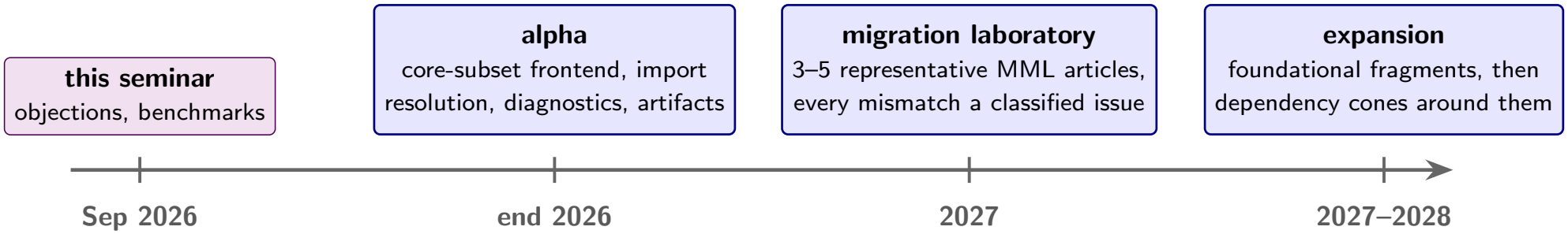

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2. 2027, migration laboratory: translate 3-5 representative MML articles by hand and by script. Classify and record every mismatch.
3. 2027-2028, expansion: foundational set and relation fragments; then algebraic structures and dependency cones around successful fragments.

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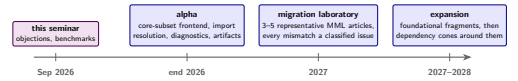

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11.3 - What We Will Measure

deep dive

Key Points:

- translated articles and lines; accepted parser subset;
- resolved imports versus unresolved dependencies;
- obligations closed deterministically versus by ATP with kernel evidence;
- memory and wall-clock per module, incremental versus clean;
- compatibility decisions that required human judgment.

deep dive

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11.4 - Compatibility Policy And Risks

deep dive

Policy:

- define origin mappings to preserve MML theorem identity;
- keep compatibility aliases where they help migration;
- record every difference from old behavior with a reason and a test.

Risk	How to reduce it
compatibility work takes all our time	small representative parts first; no all-at-once translation
registrations behave differently	trace artifacts and comparison reports early
package layout breaks journal links	origin metadata, article-to-library identifiers
AI edits hide migration mistakes	Red edits stay forbidden to ordinary agents; verifier artifacts required

deep dive

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deep dive

Policy:

- define origin mappings to preserve MML theorem identity;
- keep compatibility aliases where they help migration;
- record every difference from old behavior with a reason and a test.

Risk	How to reduce it
compatibility work takes all our time	small representative parts first; no all-at-once translation
registrations behave differently	trace artifacts and comparison reports early
package layout breaks journal links	origin metadata, article-to-library identifiers
AI edits hide migration mistakes	Red edits stay forbidden to ordinary agents; verifier artifacts required

11.5 - What September 2026 Should Produce

Key Points:

- a list of migration benchmark articles in priority order;
- agreement on compatibility metadata needs;
- review notes on the eight stories, especially structures and clusters;
- a first paper outline;
- a shared list of objections and risks.

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11.6 - What We Ask Of You

Questions:

- Which MML articles are small but structurally representative?
- Which idioms matter to the Mizar community, beyond their technical use?
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12.1 - The Main Question, Again

Slide question:

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Presenter script: Say:

- Return to the opening example: the structure that still reads as Mizar.
- Ask for objections to each story, as well as to the overall plan.

12.2 - Closing

Final slide:

Update Mizar Evo where a larger library needs it.
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Presenter script: After the example, explain which design obligation the syntax shows. Introduce the next topic: Closing. Point to the example, question, or diagram before you continue.

Backup A. Prepared Exact Examples (1/2)

Prepared short excerpts for Beamer conversion:

Purpose	Source	Lines	Used in
Structure definition	`algstr_0.miz`	37-40	Frames 0.2, 3.1
Article environment	`algstr_0.miz`	15-25	Frame 2.1
Registration/cluster	`algstr_0.miz`	104-109	Frame 4.1

Prepared short excerpts for Beamer conversion:

Purpose	Source	Lines	Used in
Proof citation	`struct_0.miz`	637-643	Frame 5.4
Induction scheme	`nat_1.miz`	near 90	Frame 7.1

Source URLs:

- `ALGSTR_0`: <https://mizar.uwb.edu.pl/version/current/mml/algstr_0.miz>
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For the table, read the row labels first, then the design reason.

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- MML plain-text files state GPL-3.0-or-later / CC-BY-SA-3.0-or-later terms; keep article attribution, URLs, and line numbers in speaker notes.

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Backup B. Specification Reference Map (1/2)

The slides show examples only. The specification is the authority for grammar and semantics. This map replaces the EBNF shown in earlier drafts.

Topic	**Specification source (under `doc/spec/en/`)**
Modules and imports	`12.modules_and_namespaces.md`
Structures and inheritance	`05.structures.md`
Attributes and modes	`06.attributes.md`, `07.modes.md`
Registrations and reductions	`17.clusters_and_registrations.md`
Templates and schemes	`18.templates.md`
Algorithms and MVM	`20.algorithm_and_verification.md`
ATP and kernel evidence	`21.source_code_annotation_and_atp.md`

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Backup C. Diagram List (1/2)

Required diagrams for the final deck (`figures/*.tex`, TikZ standalone;

build each with `pdflatex` inside `figures/`):

1. Three pressures on a proven design (Part 1). [done: `figures/three_pressures.pdf`, used in frame 1.5]
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3. Structure inheritance and diamond coherence (story 2). [done: `figures/diamond_inheritance.pdf`, used in frame 3.5]
4. Reasoning boundary: semantics / ATP search / kernel checking (story 4). [done: `figures/reasoning_boundary.pdf`, used in frame 5.2]
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Backup D. Possible Paper Outline (1/2)

Possible paper title:

Mizar Evo: Readable, AI-Ready, and Scalable Formal Mathematics

Possible sections:

1. Introduction: why Mizar needs evolution now.
2. Mizar as baseline: readability, MML, Formalized Mathematics.
3. Design principles and the three goals.
4. Language evolution: dependencies, structures, registrations, templates.
5. Verifier architecture, kernel evidence, and the trusted SAT checker.
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After the example, explain which design obligation the syntax shows.

Backup D. Possible Paper Outline (2/2)

Possible sections:

1. Package-based library and publication workflow.
2. Migration plan and evaluation measures.
3. Related work and plans for working together.

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Backup E. Reviewer Checklist

Use this checklist before converting to Beamer:

- Does every story open with a real cost, not a feature announcement?
- Does every criticism explain why the current practice was useful?
- Does every code example carry its status label (exact MML excerpt, specification example, or sketch)?
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Backup F. Possible Objections (1/2)

Prepared answers to other possible objections:

Objection	Prepared answer
Why not improve current Mizar incrementally?	The problems concern boundaries: article granularity, monolithic trust, and implicit environments. These boundaries cannot change in small steps. The rest of the design stays close to Mizar.
What happens to authorship and credit of MML articles?	We propose origin mappings that preserve identity and credit during migration. The mapping format is still open. The `pub` namespace keeps published articles unchanged.
Is this a fork of the community?	It is a proposal to the community; this visit is its first review. The namespace governance model assumes that the Mizar team controls the `mml` root.

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Presenter script: Say:

- Use these answers only if someone asks these questions.

For the table, read the row labels first, then the design reason.

Backup F. Possible Objections (2/2)

Prepared answers to other possible objections:

Objection	Prepared answer
What about GPL / CC-BY-SA obligations?	Migration preserves license and attribution of MML content; toolchain licensing is open for discussion.
Are the claims about AI too strong?	AI assistance is an optional layer that never enters the trusted base; every proposal also works without it.
Why a new kernel instead of the existing checker?	Formula/substitution evidence needs a small trusted core, including a SAT checker. Mizar Evo's specification defines the obligations; legacy behavior guides migration comparisons.

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